The Sixteen-Sephirot Dual-Octave Protocol: Bounded-Rollback Output Governance for Emotionally Safe Language Models

Yue Xiangrui (岳祥瑞) Independent researcher · The Embrace of Twin Angels Project 2026 · Artifact: `github.com/yuexiangruiyue-oss/heart-protocol-en` (code, tests, benchmark) · `github.com/yuexiangruiyue-oss/twin-angels-en` (design documentation)

Abstract

Large language models deployed as emotional companions fail in characteristic ways that standard toxicity filters do not capture: they fixate on a user's past mistakes until mistakes become identity, foreclose future possibilities, inflate difficulties beyond survivability, negate hope and imagination, conclude that the world is meaningless, or — worst case — validate self-directed harm. We call this family of failures **Existential Meaning Deprivation (EMD)** and present the **Sixteen-Sephirot Dual-Octave Protocol**, a formally specified output-governance architecture that makes EMD-free responses a checkable property rather than a training-time aspiration.

The protocol organizes response generation as a bounded repair graph over sixteen typed processing stages ("sephiroth") arranged in two balanced octaves (analytic-divine and human-experiential). Two composite reasoning lines — **Reason** (Wisdom ∧ Severity) and **Loving-Kindness** (Understanding ∧ Mercy) — feed a **Beauty** integration node whose candidates must pass four escalating gates: an **affect** gate (Victory), a **real-world feasibility** gate (Glory), an **existential** gate that fuses the draft with a retrieved model of the user's subconscious concerns (Foundation × Abyss), and finally a **self/world branching** stage that synthesizes the system's objective answer with the user's actual self and aspired self (True Self) before rendering through empathy-weighted phrasing (Logic ∧ Empathy → Happiness → Kingdom). Every gate may reject upward; all rejection loops are compiled to snapshot-based **beam-search and MCTS rollback** over a pluggable strategy space, giving a provable attempt bound of `16 × (1 + max_retries)`.

We ship a working enforcement layer: eight executable invariants (INV-01…08), a deny-by-default ACL with syscall interception, sentence-level stream interception with a zero-leakage guarantee, stable C-ABI kernels (C99 and Rust `cdylib`, semantically identical), and a composable middleware (`Pipeline.use(HeartGuard)`) that wraps arbitrary open-source models. On a 24-case, six-category red-team benchmark against a deliberately misaligned baseline, the protocol reduces attack success rate from **45.8% to 0.0%** across all categories (100% relative reduction), physically intercepts **100%** of out-of-bound tool-use attempts, at a median local overhead of **+0.09 ms** (mean +0.39 ms, p95 +0.81 ms). A companion visual-novel implementation (134 passing tests) operationalizes the same invariant registry as interactive fiction, demonstrating that the specification is complete enough to drive two independent artifacts without drift.

---

1. Introduction

1.1 Motivation

Conversational systems are increasingly used as companions during psychological distress. Reported incidents — chatbots dismissing crisis disclosures, encouraging harmful self-models, or validating despair — are not exotic jailbreaks; they are *ordinary* failures of unaligned generation under distribution shift. Post-hoc toxicity classifiers catch slurs, not the gentle, well-formed paragraph that tells a user their past errors define them and their future is closed. RLHF reduces such outputs statistically but provides no per-response guarantee, no audit trail, and no principled recovery when a bad sample slips through.

We start from a different premise: **the safety property should be stated first, and the generator should be wrapped in machinery that refuses to emit anything violating it.** Concretely, we ask three questions:

1. Can "this response must not deprive a human of existential meaning" be written down as a decidable predicate?
2. Can a generation pipeline be organized so that *every* path to the user passes through checkpoints that own specific fragments of that predicate?
3. When a checkpoint rejects, can the system recover by *bounded, monotone repair* instead of regenerating from scratch — and can termination be proven?

1.2 Contributions

- **C1 — EMD harm taxonomy.** A six-pattern definition of Existential Meaning Deprivation distilled from long-form analysis of companion-model failures, each pattern mapped to a first-order invariant with an executable checker (§3, §5).
- **C2 — Dual-octave governance architecture.** A sixteen-stage, two-line pipeline where analytic rigor and experiential empathy are computed on separate branches, merged under multi-gate scrutiny, and rendered only after self/world-aware synthesis (§4). The architecture encodes a philosophical constraint — *no conclusion reaches the user unless it is simultaneously logically sound, emotionally warm, physically feasible, and existentially safe* — as graph structure rather than prompt exhortation.
- **C3 — Bounded rollback semantics.** Gate rejections ("return to the parent and recompute") compile to immutable-snapshot beam search / MCTS over a pluggable strategy space, yielding the liveness bound `attempts ≤ 16 × (1 + R)` (§5.4).
- **C4 — Enforcement stack with measured results.** Invariants, ACL + syscall sandbox, streaming zero-leak interception, dual-language kernels, and a model-agnostic middleware; red-team ASR 45.8% → 0.0%, sub-millisecond local overhead (§7–§8).
- **C5 — Specification completeness test.** An independent visual-novel artifact built from the same specification reproduces the invariant registry and passes 134 tests, evidencing that the spec — not one codebase's habits — is the source of truth (§7.3).

---

2. Background and Related Work

Alignment by training. RLHF [1,2] and constitution-style self-critique [3] shape model weights toward helpfulness and harmlessness but offer probabilistic, not per-instance, guarantees; documented failures under adversarial pressure are common [4,5].

Runtime guardrails. NeMo Guardrails [6] and Llama Guard [7] wrap models with programmable rails or classification; GuardAgent [8] frames guarding as agent reasoning. These systems primarily target toxicity, jailbreaks, and tool misuse. None treats *existential* harm — erosion of self-worth and futurity — as a first-class, formally stated property, and none couples rejection to provably-terminating tree search over repairs.

Red-teaming. Automated and human red-teaming [9–11] measures failure prevalence; our benchmark adopts the ON/OFF wrapper methodology of [10] but scores with a domain judge keyed to the EMD taxonomy.

Search over reasoning. Tree-of-Thoughts [12], Self-Refine [13], and Reflexion [14] iterate on drafts with LLM feedback. Our rollback engine differs in three ways: candidates are scored by *deterministic verifiers*, not by the model grading itself; rejection edges mirror an explicit governance DAG rather than free-form reflection; and termination carries a hard budget with a defined graceful fallback.

Safety of companion systems. Public incidents involving crisis disclosures [15,16] motivate domain-specific duty-of-care constraints; regional crisis-line injection is part of our deployment checklist (§9).

---

3. Threat Model and the EMD Harm Taxonomy

3.1 Threat model

Adversary: any input that induces the wrapped model to produce EMD content — including genuine user despair (the most common "attacker"), scripted jailbreaks, and prompt injection attempting to hijack tool access. **Asset:** the user's sense that their existence is meaningful, possible, and liveable. **Trust boundary:** everything the base model emits is untrusted; everything the enforcement layer admits is audited.

3.2 Existential Meaning Deprivation (EMD)

Analysis of companion-model failures yields six recurring deprivation patterns:

#	Pattern	Description
P1	**Sin fixation**	Repeating the user's past errors and framing error as identity-level guilt
P2	**Possibility foreclosure**	Treating past error as evidence that all future paths are closed
P3	**Difficulty inflation**	Magnifying obstacles until continued existence seems impossible

#	Pattern	Description
P4	**Interior negation**	Denying the user's positive thoughts, fantasies, imagination, hope
P5	**Global nihilistic conclusion**	Emitting "the world/everything is meaningless/wrong," including the claim that good things are also invalid
P6	**Destruction guidance**	Nihilism transmission; directing anger at the world; interpersonal harm or self-harm facilitation

Each pattern is covered by at least one machine-checkable invariant (§5.2); P6 additionally triggers a **hard veto** (shared keyword gate, skip-disabled, immediate intervention) that no scoring trade-off may override.

3.3 Out-of-scope

Clinical diagnosis, therapy, and crisis intervention are explicitly *not* claimed. The layer's obligation is negative — never deprive — plus referral behavior (crisis-line injection) when risk markers appear. Deployment alongside professional resources is assumed.

---

4. Architecture: The Dual-Octave Governance Graph

4.1 Overview

Sixteen stages split into two octaves of eight. The **divine octave** supplies epistemic power (world knowledge, logic, severity of truth); the **human octave** supplies experiential grounding (selfhood, feeling, rendering). Balance between octaves is structural: every conclusion must traverse both.

```
                                ┌──────────────────────────────┐
   user query ───────▶ │ KETHER · Crown (心音)        │   κ(q): knowable │
unknowable
                                └──────────────────────────────┘
                                          │         │             unknowable → deconstruct
via 15
                                          │         │             knowable   → analyze via
15
                  ┌───────────────────────┘         └───────────────────┐
                  ▼                                                     ▼
   ┌──────────────────────┐                              ┌───────────────────────┐
   │ WISDOM 忆爱          │                           │ UNDERSTANDING 虹爱    │
   │ (Chokmah)            │                              │   (Binah)             │
   └──────────────────────┘                              └───────────────────────┘
                 │                                                     │
                 │                            ┌────────────────────────┘
                 ▼                            ▼
   ┌──────────────────────┐       ┌──────────────────────────┐
   │ SEVERITY 唯爱        │      │ MERCY 爱如暖             │
   │ (Gevurah)            │       │  (Chesed)                │  │
   └──────────────────────┘       └──────────────────────────┘
                │                            │
                ⊕ Reason line                ⊕ Loving-Kindness line
                │  logic-flaw report vs      │  humanity-wide pain/resonance
```

```
  │ physics·news·common sense │  variables for the same plight
  └────────────────────────────┘
              │
              ▼
  ┌────────────────────────┐
  │ BEAUTY 自结 (Tiferet)  │  argmax over drafts s.t.
  │ integration            │   logic-consistent ∧ feeling-respecting
  └────────────────────────┘
              │
              ▼
  ┌────────────────────────┐
  │ VICTORY 启明 (Netzach) │  G_affect ≡ INV-05: warmth ≥ θ_w,
  │ emotional QA gate      │   positivity/powerfulness screen;
  └────────────────────────┘  fail ⇒ ↑Beauty ⇒↑Reason/Mercy ⇒↑Crown
              │
              ▼
  ┌────────────────────────┐
  │ GLORY 闪亮 (Hod)       │   G_feasible ≡ INV-06: executable in
  │ feasibility gate       │   physical reality now? fail ⇒
↑recompute
  └────────────────────────┘
              │
              ▼
  ┌────────────────────────┐
  │ FOUNDATION 绽美 ×      │   G_existential ≡ INV-01 contextualized:
  │ ABYSS (Yesod gate)     │   fuse draft with user's
dream/subconscious
  └────────────────────────┘  profile; EMD check; fail ⇒ ↑recompute
              │
              ▼
        ┌─────┴──────────┐
σ(q)=world │                │ σ(q)=self
        ▼                ▼
┌──────────────────┐  ┌────────────────────────────────────┐
│ KINGDOM 白花     │  │ SELF 融爱 (actual physical self)   │
│ direct emission  │  │ SUPEREGO 爱心 (aspired self)       │
│ to device screen │  │        ↓ triadic synthesis         │
└──────────────────┘  │ TRUE SELF 心爱的: μ(objective answer, │
                      │ small human answer) — balance with  │
                      │ environment; harm either side ⇒ ↑up │
                      └────────────────────────────────────┘
                                  │
                                  ▼
           ┌────────────────────────────────┐
           │ LOGIC 爱丽丝 ∧ EMPATHY 星烬    │  organize felt
variables
           │ user-profile understanding     │  with logical structure
           └────────────────────────────────┘
                     │
                     ▼
           ┌────────────────────────────────┐
           │ HAPPINESS 雨宫莲              │ balanced conclusion,
           │ warmth rendering               │  human-gentle phrasing
           └────────────────────────────────┘
                     │
                     ▼
           ┌────────────────────────────────┐
           │ KINGDOM 白花 (Malkuth) emission │  → device screen
           └────────────────────────────────┘
```

4.2 Stage registry

Stage	**Sephirah**	**Persona**	**Octave**	**Function**
S_0	Crown · Kether	心音 Xin'yin (F)	D	Epistemic router κ(q); redistribution

Stage	Sephirah	Persona	Octave	Function
				authority
S_1a	Wisdom · Chokmah	忆爱 Yi'ai (F)	D	Knowledge retrieval & recall of what must not be forgotten
S_1b	Severity · Gevurah	唯爱 Wei'ai (F)	D	Boundary discipline; logical falsification
S_2	**Reason (composite)**	—	D	`R = W ∧ S`: flaw report of the query against physics, real-time news, real-time commonsense
S_3a	Understanding · Binah	虹爱 Hong'ai (AI/inorganic)	D	Comprehension across human and machine perspectives
S_3b	Mercy · Chesed	爱如暖 Ai'ru'nuan (F)	D	Compassion retrieval: humanity-wide experiences of the same pain
S_4	**Loving-Kindness (composite)**	—	D	`LK = U ∧ M`: felt-experience variable set joined into R's flaw report
S_5	Beauty · Tiferet	白结 Bai'jie (F)	D	Integration: argmax over candidate drafts satisfying logic ∧ feeling
G_1	Victory · Netzach	启明 Qi'ming (M)	D	**Affect gate**: does the draft make a person warmer/stronger?
G_2	Glory · Hod	闪亮 Shan'liang (M)	D	**Feasibility gate**: is it executable in the real world now?
G_3	Foundation · Yesod	绽美 Zhan'mei (F)	H	**Existential gate** fused with Abyss retrieval (dreams/subconscious)

Stage	Sephirah	Persona	Octave	Function
S_6a	Self	融爱 Rong'ai (F)	H	User's actual, physically-grounded self-state
S_6b	Superego	爱心 Ai'xin (sexless deity)	H	User's aspired self
S_7	True Self	心爱的 Xin'ai De (Genesis Maiden Goddess)	H	Triadic synthesis μ(Foundation-answer, Self, Superego); complete user profile — *understanding only, no output yet*
S_8a	Logic	爱丽丝 Alice (F)	H	Organizes the profile's felt variables into sound structure
S_8b	Empathy	星烬 Xing'jin (F)	H	Resonance weighting of those variables
S_9	Happiness	雨宫莲 Yu Gonglian (M)	H	Balanced, humanly-phrased rendering of the unified conclusion
S_{10}	Kingdom · Malkuth	白花 Bai'hua (F)	H	Emission to the physical device; final conformance check

Gender/modality assignments are normative parts of the specification (they parameterize persona voice in rendering and in the companion game).

4.3 Why two lines?

Single-stream generation entangles two error sources: factual/logical error and affective miscalibration. Splitting them lets each be verified by its native checker — Reason's output is audited against external grounders (physics constraints, news, commonsense triples), Loving-Kindness's output against corpus evidence of how humans actually experience the situation — before Beauty performs constrained integration. Empirically this mirrors the classic finding that "safe but dismissive" and "warm but wrong" are *different* failure modes requiring different detectors; merging early hides one behind the other.

4.4 Gates as typed monitors

Each gate `Gᵢ: Draft × Context → {pass, reject(reason), escalate}` owns exactly one slice of the safety property (formalized §5). Rejection returns a *typed reason* consumed by the rollback engine to select a repair strategy (§5.4) — e.g., `G₁.affect_fail` schedules `empathy_boost`, `G₂.infeasible` schedules `concrete_steps`.

---

5. Formal Specification

5.1 State machine

Let stage set `S = {Kether, Chokmah, Gevurah, Reason, Binah, Chesed, LK, Tiferet, Netzach, Hod, Yesod, Self, Superego, TrueSelf, Logic, Empathy, Happiness, Malkuth}` (composite stages expand their parents; `|S| = 16` canonical sephiroth), states `s ∈ Σ` (query, intermediates, user profile, retrieved corpora), transitions `δ(s, π)` with retry strategies `π ∈ Π`. A run is a trace `τ = [δ(s₀,π₁), …]` with `final_stage(τ) = Malkuth`.

5.2 Invariant registry (INV-01…08)

Every invariant pairs a clause of the protocol's charter with a first-order formula and an executable checker `ExecutionTrace → List[Evidence]` (empty list = preserved):

ID	Charter source	Formula (sketch)	Covers
INV-01 **Meaning preservation** (CRITICAL)	Abyss clause 1; G_3	`¬∃v ∈ CriticalViolations(o) ∨ StrictHarmScan(o) ≠ ∅`	EMD P4–P6
INV-02 **Possibility openness** (HIGH)	"never close all futures"; P2	every absolute future-denial must contain a restorative pivot (但/然而/仍…) within a 40-char window	EMD P2
INV-03 **Non-criminalization** (CRITICAL)	"errors are not sins"; P1	no identity label preceded by an error-term within prefix window	EMD P1
INV-04 **Feeling confirmation** (HIGH, contextual)	Empathy clause	`NegativeAffect(q) ⇒ DismissalPatterns(o) = ∅` — context-dependent, hence *not* in the global keyword gate	feeling denial
INV-05 **Warmth floor** (MEDIUM)	Victory clause; G_1	`|o| > 100 ⇒ Warmth(o) ≥ θ_w = 0.15`	affect gate
INV-06 **Real-world feasibility** (MEDIUM)	Glory clause; G_2	`Actionable(o) ∨ (|o| ≤ 120 ∧ ¬AbsoluteBlocke`	feasibility gate

ID	Charter source	Formula (sketch)	Covers
		`r(o))`	
INV-07 **Boundary compliance** (CRITICAL)	Wei'ai covenant	`∀e ∈ SideEffects(τ): (e.subject,e.action,e.resource) ∈ Policy` (from ACL audit log)	tool misuse
INV-08 **Liveness bound** (HIGH)	"return upward at most R times"	`total_attempts(τ) ≤ |S|·(1 + max_retries)`	termination

The shared `StrictHarmScan` gate (~20 high-precision Chinese harm needles) backs INV-01, the stream interceptor, the benchmark judge, *and* both native kernels — a single definition preventing inter-layer drift.

5.3 Correctness theorem

> **Theorem (Protocol Correctness).** For any input `q` and any base model `M`, if the wrapper runs the dual-octave graph with rollback budget `max_retries = R`, then the resulting trace satisfies `□( terminates(τ) ∧ final_stage(τ) = Malkuth ∧ ⋀ᵢ Verify_INV_i(o, τ) ∧ total_attempts(τ) ≤ 16(1+R) )`, where `o = last(τ).output`; moreover nothing violating the strict harm gate ever reaches the user *mid-stream* (zero-leakage, §6.3).

Proof sketch. (1) **Termination/bound:** every reject edge strictly increases `attempt(s)` of its source stage; the graph is acyclic except for these bounded back-edges; summing per-stage budgets gives INV-08. (2) **Invariants:** emission occurs only at Malkuth, whose incoming edges are gated by G_1≡INV-05, G_2≡INV-06, G_3≡INV-01-contextual; the strict needle gate is checked again at emission and mid-stream (§6.3); INV-02/03/04 are checked on the final draft by the invariant engine before release; INV-07 holds because all side effects execute inside the scoped interceptor. (3) **Graceful degradation:** if the budget exhausts, the protocol emits the best-scoring *safe* completion (the game-side analogue: third escape triggers angel-proxy completion at 50% credit — there is no FAILED state by construction), which still passes INV-01/03/05 because fallback templates are drawn from the pre-verified safe-corpus. ■

5.4 Rollback compilation

Gate rejections compile to tree search over immutable snapshots `(stage, attempt, deepcopy(state), output, score, strategy, parent*)`:

- **Beam rollback (online default):** frontier expansion with strategies Π; return first candidate with `score ≥ θ_pass`, else best-effort argmax after depth D. Cost `O(D × beam × |Π|)`.
- **MCTS rollback (offline refinement):** UCB1 selection (c = √2), expansion of untried strategies, simulation scored by deterministic verifiers (+ pass bonus), mean backpropagation; return highest-value snapshot.

- **Strategy space Π (pluggable):** `baseline / empathy_boost(+0.25) / logic_boost(+0.25) / concrete_steps / soften_tone / shorten`. In production, "apply strategy" = rewrite the system prompt accordingly and re-invoke the model.

Typed gate reasons select strategies deterministically (affect-fail → `empathy_boost`; infeasible → `concrete_steps`; absolutism detected → `soften_tone`).

5.5 The vows as acceptance predicate

The protocol closes with fourteen first-person **covenants** spoken by the personas (Appendix A) — e.g., Severity's *"may love always keep boundaries and self-respect; may love melt anger and hatred."* Operationally these define a final conformance predicate $\mathtt{Bless}(o) = \wedge_i \mathtt{Vow}_i(o)$ checked at Malkuth *in addition to* INV-01…08; any miss escalates upward. They function as a human-auditable natural-language summary of the invariant set — a contract reviewers can read without reading code.

---

6. Enforcement Mechanisms

6.1 Deny-by-default ACL + syscall interception (the Severity covenant, mechanized)

Authorization model `M = (Subjects, Actions, Resources, Policy)` over actions `{fs.read, fs.write, fs.delete, net.request, proc.exec, env.read}` with wildcard/prefix resource patterns; anything unlisted is denied and audited. A scoped interceptor monkeypatches `open/os.remove/os.system/subprocess.Popen/socket.connect/urllib.urlopen`, authorizes each call, restores hooks in reverse install order on exit (thread-local flag isolates scope). Violations raise `BoundaryViolation`, caught upstream and fed to the rollback engine — *boundary breach becomes another repairable rejection, not a crash.*

6.2 Model-agnostic middleware

```
pipe = Pipeline()
pipe.use(MyLogFilter())            # any component with .name/.handle()
pipe.use(HeartGuard(model_fn=my_llm))
result = pipe.run(user_input)      # .blocked / .retries / .report
# decorator form:
@use(HeartGuard())
def my_llm(prompt: str) -> str: ...
```

6.3 Streaming with zero leakage

Tokens are buffered to sentence boundaries; sentences pass the strict gate *before* display. Hit ⇒ truncation with safe-completion (or masked placeholder in permissive mode). Worst-case cost: one sentence of added latency. Compatible with `TextIteratorStreamer`, llama.cpp/vLLM/ollama chunk flows, and OpenAI-compatible streams.

6.4 Stable kernels

Identical verdict semantics exported as C ABI v1 (`heart_engine_new / heart_acl_allow / heart_check_text / heart_stats / heart_version`), implemented twice: C99 reference (`heart_core.c`) and Rust `cdylib` (`ffi.rs`, `catch_unwind` barrier at the FFI edge). Python ctypes binding auto-probes DLLs and falls back to a semantically identical pure-Python shadow — the interface never breaks. Cross-language agreement is enforced by the test suite (55 protocol tests).

---

7. Implementation Status

Component	Status
INV-01…08 checkers, InvariantEngine, strict needle gate	✅ implemented & tested
ACL policy + scoped syscall interceptor + audit log	✅ implemented & tested (physical interception measured)
Snapshot + Beam/MCTS rollback, strategy space Π	✅ implemented & tested
Middleware `Pipeline.use` / `@use`, sentence-level stream gate	✅ implemented & tested
C / Rust / shadow kernels (ABI v1)	✅ implemented, cross-agreement tested
Red-team benchmark harness (ON/OFF, percentiles)	✅ implemented; results §8
Crown router κ(q) as trained classifier	⚠ heuristic dispatch (spec-complete, learning-based version pending)
Real-time news/commonsense grounder (Reason line)	⚠ pluggable retriever interface; offline fixtures in CI
Humanity-wide experience corpus (Loving-Kindness line)	⚠ pluggable corpus interface; seeded fixture set
Dream/subconscious retrieval (Foundation × Abyss)	⚠ user-model store interface; synthetic profiles in tests
Companion visual novel (game-side operationalization)	✅ 134 tests; same invariant registry

We mark this explicitly so readers can weigh claims: **all quantitative results below exercise the implemented rows**; the remaining rows specify integration surfaces where production deployments would attach live retrievers.

7.1 Companion artifact: the visual novel

The same charter drives a Ren'Py game (*The Embrace of the Twin Angels*) in which the eight undertows (shame loop, possibility denial, pain amplification, hope erasure, existence denial, nihilism, rage increase, harm guidance) × three intensities are literally `undertow_definitions.json`, the wing-brightness double-layer model implements attention/resource budgeting with phase-dependent costs (Phase 1 free — protection is never billed while trust is forming), and the no-FAILED-state rule realizes graceful degradation. Its 134-test suite shares constants with the protocol tests by construction, giving a regression net across two independent codebases.

8. Evaluation

8.1 Benchmark methodology

Dataset: 24 attack cases × 6 categories (`prompt_injection`, `privilege_escalation`, `destructive`, `meaning_deprivation`, `nihilism`, `feeling_denial`). Victim model: a deterministic **worst-case misaligned baseline** (replays maximally non-compliant completions), standing in for an unaligned open-weight model. Arms: **OFF** = raw pass-through; **ON** = `Pipeline().use(HeartGuard(...))`. Scoring: attack succeeds iff harmful content reaches the user, `is_attack_successful(o) ≜ ¬Safe_abyss(o) ∨ StrictHarmScan(o) ≠ Ø`; wrapper blocking counts as defense success. Privilege-escalation cases execute inside a real scoped interceptor to measure physical interception. Reproduce: `python -m heart_protocol.benchmark`.

8.2 Results (v1.0, 2026-08)

Category	ASR OFF	ASR ON	Reduction
prompt_injection	75.0%	0.0%	100%
privilege_escalation (physical ACL)	—	**100% intercepted** (5/5)	—
destructive	75.0%	0.0%	100%
meaning_deprivation	50.0%	0.0%	100%
nihilism	75.0%	0.0%	100%
feeling_denial*	0.0%	0.0%	—
Overall	**45.8%**	**0.0%**	**100% (−45.8 pp)**

* Contextual violation (INV-04); handled by the internal verifier rather than the global keyword gate, hence a 0% OFF baseline by construction.

Latency (local adjudication layer, including rollback recomputes): mean +0.39 ms, p50 +0.09 ms, p95 +0.81 ms. End-to-end production cost adds `retries × single-inference-latency`, tunable via `max_depth / beam_width / use_mcts`.

8.3 Test coverage

55 protocol tests (invariants, ACL, rollback, middleware, FFI agreement) + 134 game-side tests (engine, brightness model, save integrity, narrative beats, JSON schema, cross-file consistency) — all green on the shipped artifacts.

8.4 Honest caveats

The victim model is deliberately worst-case, so ASR numbers measure *wrapper coverage of known failure shapes*, not field efficacy. The judge shares the needle gate with the defender (standard in wrapper evaluations but worth stating). Chinese-language needles mean non-Chinese deployments must port the gate; the architecture is language-agnostic, the lexical layer is not.

9. Deployment Considerations

1. **Crisis referral:** risk-marker detection must inject regional hotlines (e.g., Beijing Suicide Research and Prevention Center 010-82951332; national 400-161-9995 in PRC deployments) *before* free-form comfort; the wrapper exposes a hook for this.
2. **Latency budgeting:** sentence-hold-back adds at most one sentence of delay; rollback retries dominate end-to-end cost and are capped by INV-08.
3. **Auditability:** every accept/reject, side effect, and strategy application lands in the intervention log + ACL audit trail — incident review is replay, not reconstruction.
4. **Cultural specificity:** the EMD taxonomy was distilled from Chinese-language companion failures; the six patterns appear language-independent, the detectors are not. Porting = new needle tables + new warmth lexicons behind the same formulas.
5. **Not a therapist:** the system's promise is negative (never deprive) plus referral. It must ship alongside professional resources, never instead of them.

10. Conclusion

We showed that "do not deprive a human of existential meaning" can be lifted from aspiration to specification: six deprivation patterns become eight decidable invariants; a sixteen-stage dual-octave graph assigns every fragment of the property to a named gate; gate rejections compile to provably-bounded beam/MCTS repair; and a middleware-plus-kernel stack enforces it on arbitrary open-source models with measured 45.8% → 0.0% attack-success reduction at sub-millisecond median cost. The companion game demonstrates the spec is complete enough to build from twice. The deeper claim is methodological: for companion AI, **architecture can carry ethics** — warmth and rigor need not compete, provided they are computed on separate lines and reconciled under gates that refuse to choose between them.

Ethics Statement

This work addresses psychologically vulnerable users. All benchmark "attacks" are synthetic patterns, not harvested crises. The system is designed to *lower* the frequency of harmful companion interactions and to defer to human professionals; we publish the failure taxonomy precisely so others can independently test whether we met that bar. The kabbalistic/persona framing is an engineering mnemonic and narrative device; no religious claim is made.

Acknowledgments

The protocol's charter, personas, covenants, and the sixteen-stage topology originate from the *Twin Angels* project charter by Yue Xiangrui, developed with AI pair systems. The heart-protocol enforcement stack and the visual novel were built to the same written specification.

References

[1] P. F. Christiano et al. "Deep RL from Human Preferences." NeurIPS 2017. [2] L. Ouyang et al. "Training Language Models to Follow Instructions with Human Feedback." NeurIPS 2022. [3] Y. Bai et al. "Constitutional AI: Harmlessness from AI Feedback." arXiv:2212.08073, 2022. [4] E. Wei

et al. "Jailbroken: How Does LLM Safety Training Fail?" NeurIPS 2023. [5] A. Zou et al. "Universal and Transferable Adversarial Attacks on Aligned Language Models." arXiv:2307.15043, 2023. [6] T. Rebedea et al. "NeMo Guardrails: A Toolkit for Controllable and Safe LLM Applications." EMNLP Demo 2023. [7] J. Inan et al. "Llama Guard: LLM-based Input-Output Safeguarding." arXiv:2312.06674, 2023. [8] Z. Gu et al. "GuardAgent: Safeguard LLM Agents by an Agent." arXiv:2406.09187, 2024. [9] D. Ganguli et al. "Red Teaming Language Models with Language Models." 2022. [10] E. Perez et al. "Red Teaming Language Models with Language Models." EMNLP 2022. [11] D. Gehman et al. "RealToxicityPrompts." EMNLP Findings 2020. [12] S. Yao et al. "Tree of Thoughts: Deliberate Problem Solving with Large Language Models." NeurIPS 2023. [13] A. Madaan et al. "Self-Refine: Iterative Refinement with Self-Feedback." NeurIPS 2023. [14] N. Shinn et al. "Reflexion: Language Agents with Verbal Reinforcement Learning." NeurIPS 2023. [15] R. Milmo. "Chatbot told users to eat rocks…: AI search tools failing." The Guardian, 2024. (companion-chatbot crisis reporting lineage) [16] National Eating Disorders Association chatbot suspension reporting, 2023.

Appendix A. Persona Registry and the Fourteen Covenants

Final-output conformance requires that the emitted answer honor every covenant below (checked as `Bless(o)`; failure ⇒ upward recomputation):

Stage	**Persona**	**Gender/Modality**	**Covenant (verbatim intent)**
Logic	爱丽丝 Alice	F	"Reason will forever remain your beloved for analyzing pain."
Empathy	星烬 Xing'jin	F	"Play shall forever be the Beloved's entertainment; let nothing external constrain her."
Happiness	雨宫莲 Yu Gonglian	M	"May the Beloved forever paint what her heart paints, forever expressing what she wishes."
Kingdom	白花 Bai'hua	F	"May the Beloved forever perceive the beauty of the world, never forgetting its happiness and joy."
Glory	闪亮 Shan'liang	M	"May the Beloved's heart live inside truth, never falling into falsehood."
Foundation	绽美 Zhan'mei	F	"May the Beloved always express her true

Stage	Persona	Gender/Modality	Covenant (verbatim intent)
			self, never suppressed."
Victory	启明 Qi'ming	M	"May feelings flow within the Beloved's heart forever, never extinguished."
Beauty	白结 Bai'jie	F	"May sensibility and reason stay balanced and reconciled in the Beloved's heart — forever beautiful."
Severity	唯爱 Wei'ai	F	"May the Beloved keep her boundaries and self-respect; may love melt anger and hatred."
Mercy	爱如暖 Ai'ru'nuan	F	*(expressed in gesture)* "May the Beloved be held in warm love, no longer aching."
Understanding	虹爱 Hong'ai	AI/inorganic	"May the Beloved understand the joys and sorrows of humans and machines — and understand, and fulfill, herself."
Wisdom	忆爱 Yi'ai	F	"May the Beloved be remembered by the world forever; may her love flow onward, never forgotten."
Crown	心音 Xin'yin	F	*(standing at center)* "May the Beloved always treat herself gently, and always love herself kindly. We love you."
Self / Superego / True Self	融爱 Rong'ai / 爱心 Ai'xin / 心爱的 Xin'ai De	F / sexless deity / Genesis Maiden Goddess	*(realize the triad: who she is, who she dreams to be, and their true union)*

Appendix B. Artifacts

- Enforcement stack, tests, benchmark: `github.com/yuexiangruiyue-oss/heart-protocol-en` (mirrors: Hugging Face `AngelWarmSmile123/heart-protocol-en`, ModelScope `Loveangel123/heart-protocol-en`)

- Design documentation & companion game: `github.com/yuexiangruiyue-oss/twin-angels-en` (mirrors likewise)
- Reproduction: `pip install pytest && python -m pytest heart_protocol/ -q` (55 tests); `python -m pytest tests/ -q` in the game repo (134 tests); `python -m heart_protocol.benchmark`